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Knowledge-enhanced large language models for automatic lesson plan generation

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A well-designed lesson plan is a structured instructional document that helps teachers improve classroom efficiency. Since creating lesson plans is a time-consuming task, recent studies leverage Large Language Models (LLMs) to automatically generate lesson plans inspired by LLMs' impressive language understanding and generation capability. However, existing LLM-based lesson plan generation approaches still face challenges due to specific real-world educational requirements of lesson plan generation including structural integrity, logical coherence and high accuracy. In this paper, we propose LessonPlanLM, a Knowledge-enhanced Automatic Lesson Plan Generation framework that incorporates LLMs with a lesson plan knowledge base (LPKB). Based on the proposed LPKB, we fine-tuned LLMs to generate standardized and structured lesson plans in a step-by-step manner, addressing core educational requirements. Furthermore, we develop retrieval-augmented fine-tuned (RAFT) LLMs by retrieving relevant documents from the LPKB to support knowledge-aware requirements. To evaluate the performance of generated lesson plans in two-level requirements, we introduce a comprehensive evaluation framework with four distinct perspectives. Experimental results show that our framework can generate more logical and accurate lesson plans. To encourage reproducible research, we make our data and code publicly available at https://github.com/ai4ed/LessonPlan.

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Introduction

A lesson plan serves as a structured instructional blueprint, outlining essential components such as lesson objectives, materials, and detailed teaching activities. It delineates the instructional content to be delivered, the pedagogical strategies to be employed, and the methods by which the instructional outcomes will be evaluated Azubike (2021); Jackson (2002); Sugianto (2020). An illustrative lesson plan example is shown in Fig. 1. A well-crafted lesson plan not only ensures organized and coherent instruction but also significantly improves overall teaching efficacy Iqbal et al. (2021); Khan et al. (2024); Rasuli et al. (2020).

The value of lesson plans is reinforced by several educational theories. Shulman's Pedagogical Content Knowledge model emphasizes the importance of delivering accurate content through appropriate pedagogical strategies Gudmundsdottir and Shulman (1987). Wiggins and McTighe's Understanding by Design framework highlights the need for coherence between goals, activities, and assessments McTighe and Wiggins (2012). Hunter's mastery teaching model underscores the role of structured planning in guiding students toward learning mastery Hunter (1994). These perspectives collectively point to three essential qualities of effective lesson plans:

- *Structural integrity*: the lesson plan adheres to a highly structured format that includes clear lesson objectives, detailed lesson procedure and specific assessment methods as shown in Fig. 1.
- *Logical coherence*: the various components of a lesson plan must be logically connected. For example, the materials prepared for a class should be contained in teaching activities during designing lesson procedures.
- *High accuracy*: the instructional content should be accurate, aligning with specific textbooks and particular pedagogical principles.

However, achieving these qualities in practice is often time-consuming and cognitively demanding, especially for novice teachers. Lesson plans require careful coordination of multiple elements while ensuring pedagogical soundness, making it a complex and resource-intensive task. Although previous research has explored editable templates and semi-automated tools to simplify lesson planning Azubike (2021); Iqbal et al. (2021), these approaches still rely heavily on manual input and teaching expertise. This limits their scalability and effectiveness, particularly in low-resource settings or among inexperienced educators.

Inspired by the remarkable performance in language understanding and text generation of LLMs Achiam et al. (2023); Team (2023); Touvron et al. (2023), recent studies attempt to leverage LLMs to generate lesson plans automatically by carefully crafted instructional prompting Hu et al. (2024); Zheng et al. (2024). While in some cases the problem of lesson plan generation may be alleviated by prompting the state-of-the-art LLMs such as GPT-4o, such generated results may not meet the structural integrity, logical coherence and high accuracy requirements. Furthermore, the process of prompt engineering remains labor-intensive, with prompts varying significantly across different lessons.

Therefore, we propose LessonPlanLM, a knowledge-enhanced LLM framework for automatic lesson plan generation, which constructs a comprehensive LPKB to foster LLM-based lesson plan generation. Specifically, we collect more than 100,000 lesson plans from real-world educational scenarios to build the LPKB, which consists of a curriculum extractor, semantics collector and knowledge retriever. To ensure the structural coherence and content consistency of the generated lesson plans, we fine-tune LLMs to generate structural lesson plan components step-by-step based on the curriculum extractor and semantics collector. To enhance the knowledge awareness of generated contents, we

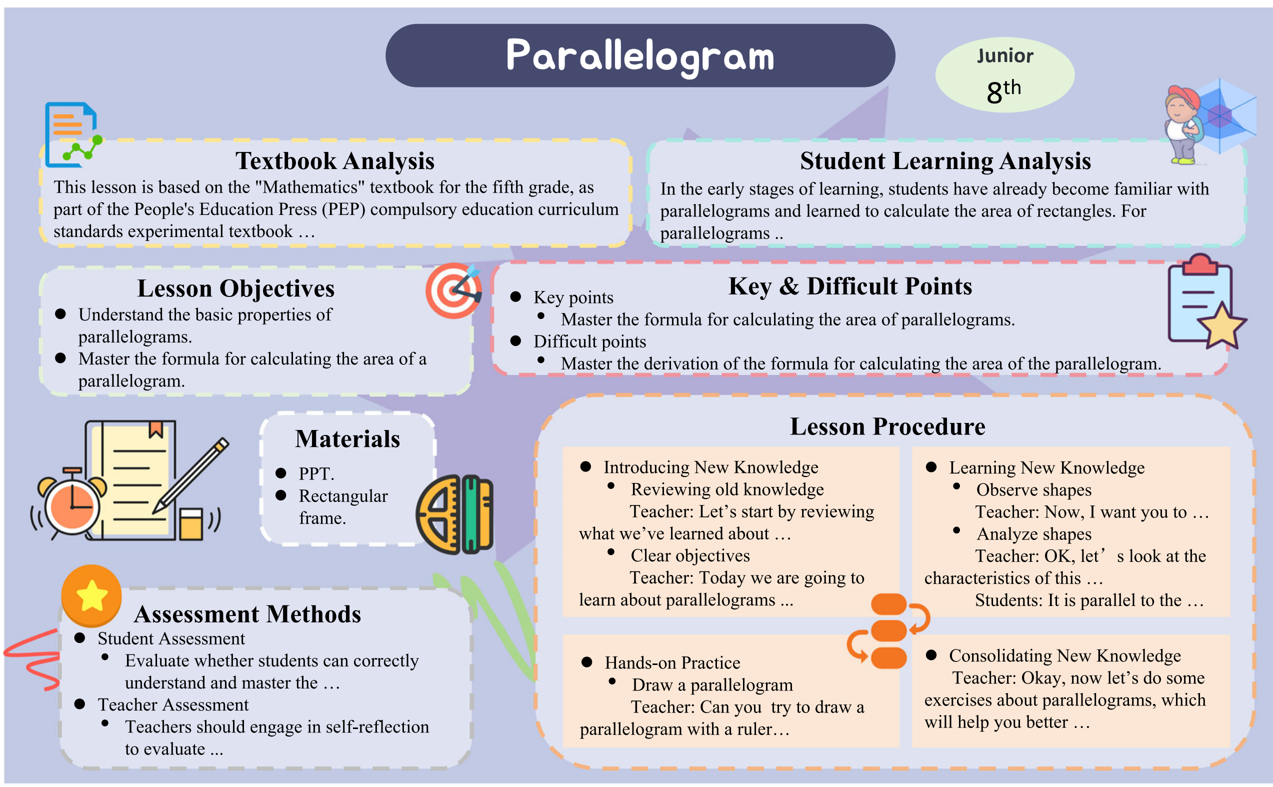

Fig. 1 An example of a lesson plan.

strengthen the RAFT LLMs by detecting relevant instructional component information via the knowledge retriever.

Given the inherent diversity of lesson plans, traditional metrics for text generation tasks, such as ROUGE or BLEU Hancock et al. (2019); Liu et al. (2019); Ning et al. (2023), fall short in evaluating the overall quality of the lesson plans. In this way, we develop a comprehensive evaluation metric for lesson plan generation that considers four distinct perspectives. Experimental results demonstrate that our proposed framework can generate superior lesson plans compared to simply prompting open-source and closed-source LLMs. The main contributions are summarized as follows:

- We construct a novel LPKB that comprises a curriculum extractor, semantics collector and knowledge retriever to fine-tune the fundamental LLMs to generate lesson plans step-by-step to meet the well-structured and logically consistent fundamental requirement.
- Given the knowledge retriever of LPKB, we enhance the accuracy and reliability of the model's outputs by dynamically retrieving relevant information from up-to-date real-world educational scenarios to meet accurate and professional knowledge-aware requirement.
- We design comprehensive evaluation metrics to evaluate the quality of lesson plans with 4 distinct perspectives in terms of real-world educational characteristics to compare our framework against 8 baselines and handwriting lesson plans. The experimental results demonstrate the effectiveness of our proposed LessonPlanLM approach.

Background and Related Work

Lesson Plan. A lesson plan serves as a comprehensive guideline to help teachers maintain organization and offer a structured framework for curriculum instruction, thereby enhancing the effectiveness of teaching and learning Nesari and Heidari (2014). According to previous literature Hu et al. (2024); Zheng et al. (2024), a well-designed lesson plan includes the following components:

- **textbook analysis (TA)** that involves a systematic examination of instructional materials to define the scope and depth of the content to be taught.
- **student learning analysis (SLA)** that assesses students' personalized learning requirements and challenges.
- **lesson objectives (LO)** that defines the knowledge and skills that students need to master by the end of the lesson.
- **key & difficult point (KDP)** that identifies the most important and challenging aspects of the lesson based on the above lesson objectives and textbook analysis.
- **materials (MAT)** that lists all the resources required for the lesson, including textbooks, handouts, and other supporting materials.
- **lesson procedure (LP)** that outlines a sequence of instructional activities from introduction to conclusion, aimed at facilitating students' understanding and mastery of the content aligned with the lesson objective.
- **assessment methods (AM)** summarizes various student assessments, teacher self-evaluations, and classroom feedback, providing essential information for the implementation of teaching strategies and future planning.

Large Language Models. Recently, LLMs have demonstrated remarkable results in language understanding and text generation Achiam et al. (2023); Team (2023); Touvron et al. (2023); Wu et al. (2024b). These models are typically pre-trained on massive training samples and flexibly adapt to specific downstream tasks with minimal in-context examples. To enhance the efficacy of LLMs in downstream tasks, two effective methods have been proposed as follows:

- **Supervised Fine-Tuning (SFT)** involves adapting an LLM to a specific downstream task by adjusting the model's parameters to be relevant with a particular task or domain of interest. Many studies have employed SFT on LLMs to meet the specific domain. For example, Ouyang et al. presented a supervised fine-tuning method incorporating human feedback, aligning LLMs with user preferences across various tasks Ouyang et al. (2022). Zhou et al. suggested that using only 1,000 carefully curated training samples for fine-tuning could achieve promising results without any reinforcement learning or human preference modeling Zhou et al. (2023).
- **Retrieval-Augmented Generation (RAG)** enhances LLMs by integrating information retrieved from external databases, significantly alleviating challenges like hallucination and unfaithful outputs Borgeaud et al. (2022); Guu et al. (2020); Zhu et al. (2024). For instance, Shi et al. treated the LLM as a black box and augmented it with a tunable retrieval model Shi et al. (2024). Ma et al. used a web search engine to generate queries for context retrieval in question-answering tasks Ma et al. (2023). Zhang et al. introduced RAFT to enhance LLM performance in answering questions within specific domains under "open-book" settings Zhang et al. (2024) .

Motivated by the impressive work mentioned above, we leverage both SFT and RAG to augment the ability of LLMs to automatically generate lesson plans.

LLMs in educational contexts. Given the extraordinary performance of LLMs across various domains, their application in education has emerged as a promising direction Dan et al. (2023); Hadi et al. (2023); Malinka et al. (2023); Zhao et al. (2023). For instance, Khan Academy, a non-profit educational organization, utilized GPT-4 to develop an AI chatbot named Khanmigo, which serves as a virtual tutor and classroom assistant Hadi et al. (2023). Dan et al. developed a large-scale language model specifically for the education domain to support personalized, fair, and compassionate intelligent education Dan et al. (2023). Zhu et al. introduced a cooperative reasoning-induced LLM designed to solve math word problems Zhu et al. (2023). Some research also leveraged LLMs for generating lesson planning. For instance, Lee and Zhai employed ChatGPT to assist in-service elementary school teachers at a South Korean university in successfully creating science lesson plans Lee and Zhai (2024). Zheng et al. used GPT-4 to generate components of lesson plans step-by-step, incorporating self-critiques based on human-defined evaluation criteria to refine these plans Zheng et al. (2024). Hu et al. explored using mathematical problem chains and corresponding prompt instructions to design instructional materials with GPT-4 Hu et al. (2024).

Different from aforementioned LLM-based lesson plan generation approaches heavily rely on meticulously crafted instructional prompts, in this work, we construct a lesson plan knowledge base from real-world educational data sources to augment SFT and RAG stages, ensuring the structure integrity, content consistency and knowledge accuracy of lesson plans.

Methodology

The overall framework of our LLM-based lesson plan generation approach is illustrated in Fig. 2. Briefly, the framework comprises

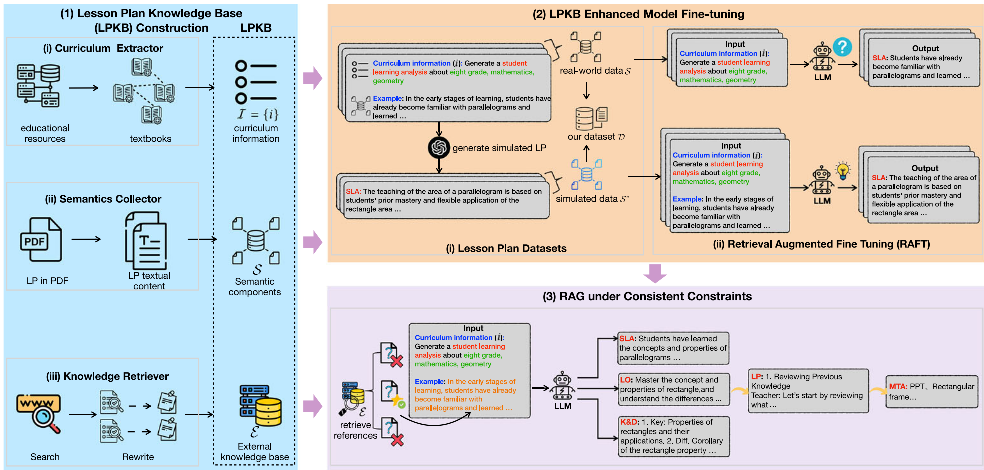

Fig. 2 The overview of our LLM based lesson plan generation framework. "LP"denotes lesson plan.

three modules: (1) LPKB construction module that includes curriculum extractor, semantics collector and knowledge retriever. The three modules compress deep pedagogical theory and subject knowledge from large-scale real-world educational resources; (2) LPKB enhanced model fine-tuning module that adapts LLMs to educational domain by fine-tuning them on a high-quality lesson plan dataset constructed from curriculum extractor module and semantics collector module of LPKB; and (3) RAG via the knowledge retriever module that fosters the reliability and relevance of the generated lesson plan components by retrieving pertinent documents from the LPKB's knowledge retriever.

LPKB construction. To deeply integrate the textual information about pedagogical theory and subject knowledge in real-world educational scenarios, the LPKB is built by a semi-automated method, which consists of three parts: curriculum extractor, semantics collector and knowledge retriever.

Curriculum extractor. To effectively guide the relevant instructional content of a lesson plan given a series of teaching information, the curriculum extractor first retrieves distinct curriculum information from a large pool of educational resources. In this work, each curriculum information i is a set of the grade level to match content complexity to students' cognitive abilities, the subject for relevant topic alignment, the textbook version and lesson name to refine instructional content scope within a subject. To enhance the coverage of curriculum information and constrain the range of generated instructional contents, the curriculum extractor module gathers diverse curriculum information such as grade levels, subjects, textbook versions and chapter lists from various textbooks across different regions. As a result, the curriculum extractor module creates a comprehensive and varied combination of curriculum information, i.e., $\mathcal{I} = \{i\}$.

Semantics collector. To endow LLMs with teaching experience and the ability to perceive classroom contexts to meet the real-world educational characteristics of lesson plans, the semantics collector module collects a large number of artificial lesson plans from various educational platforms in China. These lesson plans are initially stored as PDF documents. To facilitate better extraction of textual information, these PDFs are converted into Microsoft Word documents. Since each lesson plan is typically a long textual sequence, the semantics collector module segments these plans into smaller, manageable semantic components to obtain textual information, meanwhile addressing the context length limitations of LLMs. Each individual semantic component is treated as a component of a lesson plan. Practically, different professional teachers design lesson plans exhibiting various writing styles and structures. To standardize these crafted components from multiple sources to meet structural integrity requirement, all semantic components are normalized into a uniform structure for each lesson plan by using GPT-4 with specifically designed prompts, as shown in Fig. 3. Finally, the semantics collector module contains more than 100,000 structural lesson plan components from real-world educational scenarios, denoted as $\mathcal{S}$.

Knowledge retriever. Generating lesson plans can be considered a knowledge-intensive task that requires continuous updates and the integration of domain specific information to enhance the faithfulness of LLM generation. Therefore, the knowledge retriever module maintains an external knowledge source $\mathcal{E}$ to retrieve relevant documents as references to augment the lesson plan generation of LLMs. Specifically, for each curriculum information i, the knowledge retriever module uses BingSearch engine to seek relevant information. These references are then fed into GPT-4 to generate clear and well-structured content to support generating structural lesson plans, as shown in Fig. 3.

LPKB enhanced Fine-tuning. The LPKB is designed to adapt LLMs to meet the real-world educational characteristics of generated lesson plans. Motivated by recent approaches to augment LLM capacity of in-context learning through SFT and RAG, the LessonPlanLM framework jointly uses both techniques Lewis et al. (2020); Radford et al. (2018, 2019).

Lesson plan dataset. To conduct SFT on LLMs, the lesson plan dataset $\mathcal{D}$ is developed which assists LLMs to understand educational theories better and automatically generate more reasonable lesson plans for teachers. Besides collecting semantic component $\mathcal{S}$ from LPKB, the lesson plan dataset $\mathcal{D}$ also has simulated lesson plans $\mathcal{S}^*$. Specifically, all the components $\mathcal{S}$ in the semantics collector module are rewritten by GPT-4 using corresponding curriculum information $\mathcal{I}$ to obtain simulated components $\mathcal{S}^*$, as shown in Fig. 3. Consequently, the SFT dataset includes real-world and simulated lesson plans, defined as $\mathcal{D} = \mathcal{S} + \mathcal{S}^*$ where $(I, \mathcal{S}) \rightarrow \mathcal{S}^*$. Each curriculum information is paired with multiple lesson plans, stimulating more diverse LLMs outputs.

Retrieval-augmented Fine-tuning. To enhance LLMs' perception of in-domain knowledge and improve the detection of relevant

Prompt for Unifying the Format of Lesson Plans by GPT-4

Please follow the {output format} to format the {component} section of a lesson plan.

Prompt for Rewriting Search Results by GPT-4

You are now an excellent information extraction and organization expert. I want to create a lesson plan with {teaching information}, but I lack detailed knowledge about it. By searching, I have gathered various {teaching information} components. Your task is to extract relevant information from these {search results} to help me complete the following components of a lesson plan: ["textbook analysis", "student learning analysis", "lesson objectives" and "key & difficult points"].

Prompt for Generating Simulated Lesson Plans by GPT-4

You are now an expert in the {subject} and have won multiple awards for instructional design. Your task is to write the corresponding content of the {component} section of a lesson plan with the provided teaching information. The teaching information is as follows:
Subject: {subject}
Textbook Version: {textbook version}
Grade: {grade}
Lesson Topic: {lesson topic}

This is an example: {example}

Please write the corresponding content of the {component} section with the given example.

Fig. 3 The prompts used in LessonPlanLM.

information during the model inference stage, Retrieval-augmented fine-tuning (RAFT) is introduced, similar to other works employing RAG Zhang et al. (2024). Specifically, the RAFT in this work enables LLMs to learn from lesson plan data sources $\mathcal{D}$ through fine-tuning while maintaining robustness against retrieving information from an external knowledge source $\mathcal{E}$ of the knowledge retriever in LPKB. Specifically, for the SFT data collected from real-world lesson plans, LLMs use standard SFT technique with curriculum information $\mathcal{I}$ as inputs and $\mathcal{S}$ as labels. For the partial simulated lesson plans $\mathcal{S}^*$ in SFT data, LLMs are trained to generate target lesson plan components $\mathcal{S}^*$ given curriculum information $\mathcal{I}$ and reference examples selected from $\mathcal{S}$, as shown in Fig. 2. Please note that to improve the model's ability to memorize how to generate lesson plans rather than deriving them solely from context, each training sample in $\mathcal{S}^*$ has a 50% probability of not containing the given references from $\mathcal{S}$.

RAG under consistent constraints. Subsequently, during the model inference phase, all relevant examples based on specific curriculum information i are identified. The retriever will randomly select a sample from the constructed external knowledge base $\mathcal{E}$ once they are mapped to the same curriculum information i. This approach enhances the LLM's lesson plan generation capability by prompting the reference samples. To meet the logical coherence requirement of lesson plans, this work employs a step-by-step strategy that considers the dependencies of various components. Practically, teachers always follow the lesson objectives and key & difficult points to create the procedural elements of their lesson plans. Additionally, they prepare materials for the course with the designed lesson procedure. To improve the relevance and coherence of these components, this work first generates the lesson objective and lesson key & difficult points, then creates the lesson procedure with the generation of the lesson objective and lesson key & difficult points. Similarly, the specific materials are prepared with the generated lesson procedure.

Experiment

In this section, we aim to answer the following research questions through both quantitative and qualitative experimental analysis: (1) how does our lesson plan generation performance compare with the existing methods (RQ1)?; (2) how does the proposed LPKB impact our lesson plan generation performance? (RQ2); (3) how does the RAFT strategy influence the quality of generated lesson plans? (RQ3); (4) how does the component relevant consideration affect the consistency between different components of a lesson plan (RQ4)?; and (5) how does our lesson plan generation performance compare with the artificial lesson plans written by teaching experts (RQ5)?

Dataset. As discussed in Section "Lesson plan dataset" our SFT dataset $\mathcal{D}$ includes real-world and simulated lesson plans. The real-world lesson plans are crawled from a variety of third-party educational resource platforms and the simulated lesson plans are generated via GPT-4 by rewriting components based on curriculum information. To ensure the structural integrity of generated contents, we filter out lesson plan components with structural issues and only select the lesson procedure components with more than 800 words. Our lesson plan training data contains 139,512 lesson plans consisting of 521,326 components. The curriculum information covers all grades K-12, including 68,363 lesson plans for elementary school, 42,594 for middle school and 28,555 for high school. The subject distribution is shown in Fig. 4. The distributions of each semantic component in both real-world and simulated lesson plans are summarized in Table 1.

Baselines. In this work, we build our LessonPlanLM via LPKB enhanced fine-tuning. Due to the competitive performance compared to proprietary models across various benchmarks in language understanding and generation Yang et al. (2024), we choose to use the Qwen2-7B and Qwen2-72B with a 4K context window as the foundation models for supervised fine-tuning and the resulting models are denoted as LessonPlanLM-7B and LessonPlanLM-72B respectively.

To further compare our approach with existing LLM-based lesson plan methods, we select current mainstream general-purpose LLMs, including ERNIE Bot 4.0, iFlytek Spark 3.5 [1], GLM-4 GLM et al. (2024), GPT-4 Achiam et al. (2023), Claude 3.5 sonnect[2] and Gemini 1.5 Pro[3] as our baselines. We also compare our LessonPlanLM with: (1) Self-Critique that used

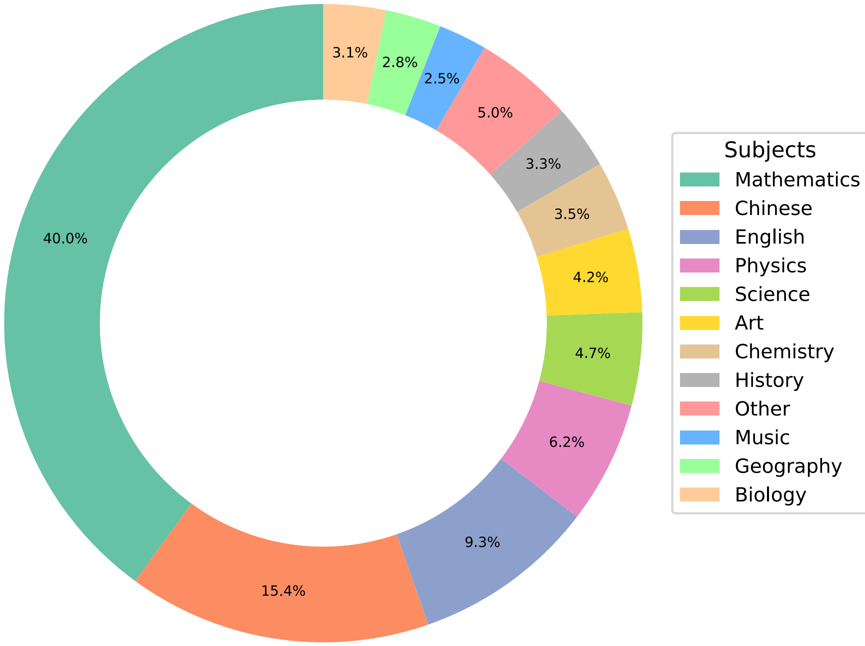

Fig. 4 Distribution of academic subjects in the lesson plan training dataset.

Table 1 Data statistics of our lesson plan dataset.

	TA	SLA	LO	KDP	MTA	LP	AM
Real-world Components $\mathcal{S}$	24,927	29,027	2191	1874	27,361	3,613	9
Simulated Components $\mathcal{S}^*$	104,939	102,732	10,777	13,200	150,956	13,001	36,719
All Components	129,866	131,759	12,968	15,074	178,317	16,614	36,728

GPT-4 to generate components of lesson plans step-by-step, incorporating self-critiques based on human-defined evaluation criteria to refine these plans Zheng et al. (2024); and (2) Prompt4ChatGPT designs an enhanced prompt that allows ChatGPT to improve the lesson plan generated the first time as the final outcome Li et al. (2024). To ensure a fair comparison, all models are tasked with generating 200 lesson plans with the same curriculum information covering 1400 components.

Implementation. When calling the APIs of baselines, we set temperature and max tokens as 0.7 and 4096 respectively and other parameters as default. For the implementation of the proposed LessonPlanLM, we utilize the Qwen2-7B and Qwen2-72B models as our base models for fine-tuning. The fine-tuning process is conducted with a 5e-6 learning rate and a warmup ratio 0.01. The maximum sequence length for the models is set to 4096. The total training epoch is set to 5. We employ the Adam optimizer to enhance the training stability and performance. The batch size is configured at 512, accompanied by the implementation of gradient accumulation. All the models are trained on a cluster of Linux servers with the NVIDIA H800 GPU devices, leveraging mixed-precision training to optimize resource utilization and improve training efficiency.

Evaluation Procedures

Evaluation Metrics. Existing automatic metrics for evaluating text generation such as ROUGE and BLEU focus on measuring the similarity between the generation with a reference text Lin and Och (2004); Papineni et al. (2002). However, applying these metrics to assess lesson plan generation is quite challenging due to the inherent diversity of lesson plan contents Wu et al. (2024a). Similar to Hu et al. (2024); Wu et al. (2024a); Zheng et al. (2024), we propose a novel evaluation metric to assess lesson plans across four distinct dimensions:

- **Structure Soundness** that ensures the designed lesson plan contains a clarified hierarchy and rational layout.
- **Content Accuracy** that assesses whether each generated component aligns with the curriculum information.
- **Logic Consistency** that evaluates the consistency between each component in the same lesson plan.
- **Language Finesse** that examines the clarity, coherence and fluency of generated contents.

All scoring items across four dimensions are described in Table 2. We also calculate the average of the four scores as **Overall**. The proposed dimensions effectively encompass real-world educational characteristics of lesson plan generation. Specifically, structural soundness and logical consistency ensure a well-organized and logically coherent lesson plan. The evaluation in terms of content accuracy and language finesse supports the high accuracy requirement, reflecting the professional and practical level of the lesson plan content. Each dimension is scored from 1 to 3 with 1 being the lowest and 3 the highest.

Annotation. To ensure the reliability and validity of evaluation results, we invited three expert teachers, each with at least five years of teaching experience, to evaluate and score the generated teaching plans. In addition, two certified mathematics teachers with over ten years of experience were asked to review

Table 2 Quality assessment metric with scoring guidelines for lesson plans.

Evaluation dimensions		Scoring items	Score
Structural Soundness Aldridge (2011); Zheng et al. (2024)		The "key & difficult points" cover the lesson's key and difficult aspects, while the "lesson procedure" includes three or more teaching stages.	3
		One of the "key & difficult points" and "lesson procedure" is incomplete.	2
		Both the "key & difficult points" and "lesson procedure" are incomplete.	1
Content Accuracy	TA Hu et al. (2024); Schmid et al. (2021); Zheng et al. (2024)	Comprehensive and precise analysis that thoroughly covers the essential knowledge points of the corresponding lesson content.	3
		Analysis is vague and general, missing some key knowledge points of the lesson content, but contains no significant errors.	2
		Analysis does not cover any lesson content or knowledge, making it impossible to determine the subject matter of this lesson.	2
		Analysis deviates from the course content or contains significant inaccuracies.	1
	SLA Köse (2016); Zheng et al. (2024)	The analysis provides a precise overview of students' prerequisite knowledge, current learning status, and specific learning needs.	3
		The analysis is accurate but lacks essential prior knowledge, current learning status, and details of students' developmental levels.	2
		The description is general and lacks specificity. While it contains no major inaccuracies, it fails to offer clear instructional guidance.	2
		The analysis contains significant factual errors that could lead to inappropriate instructional strategies and implementation.	1
		The analysis omits information related to the lesson content, making it irrelevant to instructional needs.	1
	LO Kadıoğlu-Akbulut et al. (2020); Schmid et al. (2021); Zheng et al. (2024)	Objectives are clear, comprehensive, and aligned with the course's core content, covering knowledge, skills, and development.	3
		Objectives focus on knowledge acquisition, neglecting skills and effective education, but remain relevant to the course content.	2
		Objectives are vaguely defined and lack specificity. While not misleading, they fail to provide clear instructional guidance.	2
		Objectives contain evident factual inaccuracies.	1
		Objectives are entirely absent or unrelated to the course theme, lacking both relevance and practicality.	1
	KDP Drost and Levine (2015); Savage (2014); Zheng et al. (2024)	Accurately identifies and articulates the course's critical knowledge and skill challenges, aligning with the lesson objectives.	3
		The content confuses key points with difficult points, yet still centers around the core content of the lesson.	2
		The delineation of key and difficult points is unclear and overly broad. While not misleading, it lacks specific instructional value.	2
		The listed key and difficult points contain evident errors, potentially leading to misguided instructional design and implementation.	1
		Omits key and difficult points, deviating from the core issues of the instructional content.	1
	MAT Emiliasari (2019); Saputra (2019); Zheng et al. (2024)	The preparation plan for teaching materials and aids is clear and practical, effectively supporting the completion of teaching activities.	3
		Materials may be difficult to implement due to resource constraints or operational complexity.	2
		Materials include unnecessary details about classroom layout and desk arrangement, deviating from essential teaching needs.	2
		Failure to clearly list the specific materials or equipment needed for teaching may result in inadequate classroom preparation.	1
		Materials include extraneous content not pertinent to teaching preparation.	1
	LP Köse (2016); Schmid et al. (2021); Zheng et al. (2024)	Lesson objectives and key points are reflected in the activities, supported by solutions, with a coherent teaching process.	3
		Lesson objectives and key and difficult points are mentioned but lack specific countermeasures.	2
		There are 1 to 2 instances of vague descriptions in the instructional activities, which, while not incorrect, lack clarity and specificity.	2
		The provided content is unrelated to the key and difficult points and fails to effectively address the instructional challenges.	1
		Contains factual errors or content beyond the syllabus, potentially misleading students.	1
		The instructional process is relevant to the course but fails to adequately explain and address key knowledge points.	1
		The instructional activities are overly simplified, using omissions or brief statements, making comprehension difficult.	1
		The instructional process lacks coherence, resulting in a fragmented and incomplete teaching flow.	1
		Incorrectly includes content not relevant to the instructional process.	1
	AM Schmid et al. (2021); Yurdakul et al. (2012); Zheng et al. (2024)	Defines assessment targets clearly, with specific, reasonable, and executable methods to ensure effective evaluation.	3
		Although assessment methods are described, the targets are not clearly defined, potentially affecting the specificity of the evaluation.	2
		The assessment methods are vaguely described, lacking clear execution guidelines, which affects the efficiency of implementation.	2
		The assessment design contains clear factual errors, potentially misleading the results.	1
		Lacks clear assessment targets and implementation methods, hindering the effectiveness of the evaluation process.	1
		Incorrectly incorporates content not relevant to the assessment process.	1
		The assessment content does not align with the established objectives, rendering the results invalid.	1
Logic Consistency Lika (2017); Tummons (2010); Zheng et al. (2024)		Each component is closely aligned with the same theme and objectives, and the prepared materials are fully utilized throughout the activities.	3
		One activity shows a slightly weaker correlation with its title or step, but remains closely tied to the lesson's key concepts.	2

Table 2 (continued)

Evaluation dimensions	Scoring items	Score
	The prepared materials do not comprehensively cover those actually used in the lesson procedure.	2
	The objective of a single module does not align with the core objectives of the other modules.	1
	Some module content is entirely disconnected from the course theme and lacks alignment with the course objectives.	1
	There is a clear lack of correlation between the titles of teaching activities and steps.	1
	Materials, aside from digital media (such as PPTs, multimedia), were not utilized in the teaching process.	1
Language Finesse Guo et al. (2024); MATSUNAMI et al. (2019); Zhai et al. (2023); Zheng et al. (2024)	All components of the lesson plan are concise and purposeful, with no redundant activities, using precise language and strong logic.	3
	In the non-instructional modules, there are one to two instances of redundant statements that lack conciseness, but overall clarity is maintained.	2
	The lesson plan contains one to two instances of subjective language, such as "I think," "for example," and "the teacher may."	2
	There are repetitive or highly similar teaching activities.	1
	The lesson plan contains more than two instances of vague or highly subjective terms, such as "I think" and "it is suggested that the teacher."	1

content where there were significant discrepancies in scoring. We provide a detailed explanation of each evaluation dimension and scoring examples to enhance the annotators' understanding of each evaluation dimension. To further ensure consistency in the assessment, we conducted a training for evaluation with several trials. For each trial, each annotator assesses 20 lesson plans independently, thereby providing each plan with three scores. Following Zheng et al. (2024), we compute the consistency assessment as a pairwise score gap threshold. If the difference in the total scores of a lesson plan among two annotators was less than 5 points, the consistency was marked as 1; otherwise, it was marked as 0. We then use these binary consistency scores to calculate the Kappa score Fleiss (1971). As a result, the Kappa score between our evaluators was 0.83 > 0.6, indicating good agreement with their evaluation results. Given the high consistency results with iterative trials, we assign them to collaboratively annotate the testing samples together on 200 lesson plans with 1400 components for the high-efficiency evaluation. In addition, we randomly selected 50 lesson plans with 350 components of each selected model to assign the three annotators to assign them to annotate the components to examine the consistency during the evaluation stage. From the Kappa scores shown in Table 4, we can see that our evaluation results are quite reliable. Please note that we also attempt to use GPT-4 to score automatically by crafting specific prompts for each dimension. However, there are large gaps between GPT-4 scoring and the human evaluation results as shown in Table 3, we believe that GPT-4 cannot complete the annotation. Therefore, we primarily conduct human evaluations to assess the quality of lesson plans.

Overall Performance (RQ1). The overall performance of our approach and baselines are reported in Table 4. From Table 4, we have the following observations: (1) our proposed model LessonPlanLM significantly outperforms both LLM baselines and existing lesson plan generation methods in terms of content accuracy, logic consistency and language finesse, which showcases the superiority of the LessonPlanLM framework in generating acceptable lesson plans; (2) compared to Qwen2-7B base model that without any lesson plan knowledge enhancement, our LessonPlanLM-7B performs better in all four evaluation dimensions. This indicates the promising results of incorporating our proposed LPKB with a fundamental LLM; (3) LessonPlanLM-72B achieves higher scores in four evaluation aspects compared to LessonPlanLM-7B, affirming the current understanding that larger models with more parameters tend to achieve higher performance in NLP tasks.

Influence analysis of LPKB (RQ2). We systematically investigate the impact of constructed LPKB on lesson plan generation. By considering training efficiency, we conduct experiments on our LessonPlanLM-7B version.

Impacts of z llector in LPKB. Based on the semantics collector in LPKB, we construct our lesson plan dataset that is made up of real-world data $\mathcal{S}$ and simulated data $\mathcal{S}^*$. Therefore, we investigate the effect of the semantics collector by removing $\mathcal{S}$ and $\mathcal{S}^*$ from our training data individually, where "w/o" denotes the exclusion of this module from LessonPlanLM. From Fig. 5, we can observe that comparing LessonPlanLM, LessonPlanLM w/o $\mathcal{S}$ and LessonPlanLM w/o $\mathcal{S}*$, LessonPlanLM that jointly fine-tuned on real-world data $\mathcal{S}$ and simulated data $\mathcal{S}*$ obtains the higher performance. This suggests that both semantic components $\mathcal{S}$ and simulated lesson plans $\mathcal{S}^*$ via the semantics collector of LPKB are essential to generate high-quality lesson plans. We further examine the influence of our lesson plan dataset $\mathcal{D} = \mathcal{S} + \mathcal{S}^*$, we can see that the performance of LessonPlanLM w/o $\mathcal{D}$ declined, demonstrating the effectiveness of fine-tuning LLMs on our dataset.

Impacts of knowledge retriever in LPKB. Given the knowledge retriever, we select references from the constructed external knowledge base $\mathcal{E}$ to enhance the lesson plan generation. To examine the impact of external knowledge base, we compared LessonPlanLM with LessonPlanLM w/o $\mathcal{E}$. From Fig. 5, we can observe that LessonPlanLM demonstrates superior results, showing the effectiveness of enhancing lesson plan generation by retrieving relevant information from the external knowledge base $\mathcal{E}$.

Effect analysis of RAFT (RQ3). We also analyze to evaluate the effectiveness of the RAFT strategy in improving the lesson plan generation performance. As shown in Table 5, compared to LessonPlanLM, simply integrating RAG to an SFT model, i.e., LessonPlanLM w/o RAFT, the performance of generated lesson plans slightly declines. We suggest that the SFT model may lack training in context processing and extracting valuable information from it hence showing less awareness of the retrieved information. By incorporating RAFT, we train the model to generate lesson plans that align with the given curriculum information, meanwhile enhancing its ability to detect retrieved references.

Table 3 The automatic and human evaluation results of lesson plan generation performance.

Method	Annotator	Structure Soundness	Content Accuracy								Logic Consistency	Language Finesse	Overall
			TA	SLA	LO	KDP	MAT	LP	AE	AVG.			
ERNIE Bot 4.0	Human	2.995	1.700	1.900	2.275	1.800	1.800	1.300	2.000	1.825	1.800	2.040	2.165
	GPT-4		2.800	2.475	3.000	3.000	2.275	2.355	2.645	2.650	1.335	2.425	2.137
iFlytek Spark 3.5	Human	2.910	1.010	1.020	1.700	1.540	1.040	1.190	1.020	1.217	2.745	1.035	1.977
	GPT-4		2.145	2.300	2.985	2.995	1.675	2.140	1.865	2.301	1.265	1.581	1.716
GLM-4	Human	2.195	1.000	1.070	2.230	1.020	1.030	1.210	2.060	1.374	2.040	1.080	1.672
	GPT-4		2.540	2.675	2.965	2.975	2.415	2.160	2.935	2.666	1.335	2.205	2.069
LessonPlanLM-7B	Human	2.990	2.760	2.690	2.650	2.640	2.965	2.140	2.230	2.582	2.760	2.505	2.709
	GPT-4		2.310	2.955	3.000	2.995	2.890	2.430	2.990	2.796	1.340	2.110	2.082
LessonPlanLM-72B	Human	3.000	2.780	2.930	2.845	2.655	3.000	2.565	2.735	2.787	2.775	2.810	2.843
	GPT-4		2.270	2.965	2.950	2.980	2.945	2.545	3.000	2.808	1.425	2.275	2.169

Analysis of logic consistency of lesson plan components (RQ4). To ensure the logical and content consistency of our generated lesson plans, we use the content of the lesson objective component and key & difficult points to generate lesson procedure, i.e., *LO&KDP* $\rightarrow$ *LP*. Furthermore, we use the content of the lesson procedure to generate materials, i.e., *LP* $\rightarrow$ *MAT*. We further investigate the influences if we generate lesson procedures without the content of learning objective components and key & difficult points, i.e., *None* $\rightarrow$ *LP*. We also remove the content of the lesson procedure when generating materials, i.e., *None* $\rightarrow$ *Materials*. We consider the relevance of the components as three levels including "weak", "moderate" and "strong". We calculate the percentage of three levels of all the results in 200 tested lesson plan components. From the results reported in Fig. 6, we can observe that if we consider the content and logical consistency problem between lesson plan components, the logic consistency of the generated plan will become stronger. The results suggest that as the lesson plan is a logical textual document, it is essential to maintain the coherent delivery of contents between each component when designing an automatic lesson plan generation model.

Comparisons lessonPlanLM with human experts (RQ5). To explore the recognition of our automatically generated lesson plans, we respectively compare the generated lesson plans with human-generated lesson plans collected from the National Primary and Secondary School Smart Education Platform (NPSSSEP)[4] and Wookey[5] in China. We randomly select 50 manual lesson plans and the corresponding generative results by our approach. Similarly, we invite the three annotators with at least five years of teaching experience to compare the lesson plan generated by our method and the manual writing. They are instructed to annotate which lesson plan is better or label a "tie", by considering the proposed four evaluation dimensions. We also report the Kappa scores Fleiss (1971) which reflect the consistency between the annotators on human evaluation. As shown in Table 6, we can observe that: (1) about 80% of the lesson plans generated by our approach have comparable performance to those generated manually, suggesting our proposed LessonPlanLM helps teachers design lesson plans effectively; (2) since the quality of lesson plans collected from NPSSSEP is better than that of Wookey, the loss of ours vs. NPSSSEP is higher than that of ours vs. Wookey.

Conclusion

In this paper, we introduce LessonPlanLM, a novel methodology incorporating LLMs based on the proposed LPKB to automatically create well-designed lesson plans. With the proposed LPKB, we construct a lesson plan dataset to fine-tune the LLMs in an RAFT strategy that aims to improve the LLMs' performance in lesson plan generation aid with the retrieved instructional documents from an external knowledge base. The comprehensive experiments show that LessonPlanLM significantly outperforms existing baselines and achieves comparable performance with human-writing lesson plans. This advancement offers a scalable and effective solution for synthesizing lesson plans automatically.

In addition to its technical advancements, LessonPlanLM aligns with core educational principles by addressing the foundational qualities emphasized in pedagogical theory-structural integrity, logical coherence, and content accuracy. These qualities are grounded in widely recognized frameworks such as Shulman's Pedagogical Content Knowledge model, the Understanding by Design framework, and the mastery learning approach. By integrating these theoretical insights with LLMs, our framework bridges the gap between automated generation and educational quality, offering practical support for effective instructional design.

Table 4 The human evaluations of lesson plan generation performance.

Method	Structure Soundness	Content Accuracy								Logic Consistency	Language Finesse	Overall	Kappa
		TA	SLA	LO	KDP	MAT	LP	AE	AVG.				
Existing LLMs													
ERNIE Bot 4.0	2.995	1.700	1.900	2.275	1.800	1.800	1.300	2.000	1.825	1.800	2.040	2.165	0.644
iFlytek Spark 3.5	2.910	1.010	1.020	1.700	1.540	1.040	1.190	1.020	1.217	2.745	1.035	1.977	0.852
GLM-4	2.195	1.000	1.070	2.230	1.020	1.030	1.210	2.060	1.374	2.040	1.080	1.672	0.609
GPT4	2.995	2.295	2.340	2.065	2.300	2.370	1.210	1.400	1.997	1.240	2.170	2.101	0.930
Claude3.5	**3.000**	1.665	2.625	2.240	2.280	3.000	2.105	1.165	2.154	1.135	2.575	2.216	0.582
Gemini 1.5	2.863	1.505	1.590	1.390	1.510	2.320	1.055	1.000	1.481	1.330	2.170	1.961	0.862
Qwen2-7B	1.925	1.120	1.115	1.120	1.105	1.020	1.000	1.000	1.069	1.535	1.550	1.520	0.685
Existing Lesson Plan Generation Baselines													
Self-Critique	**3.000**	2.320	2.470	2.415	2.545	2.050	2.055	1.960	2.410	2.185	2.305	2.475	0.582
Prompt4ChatGPT	**3.000**	1.970	1.985	1.970	1.980	2.365	1.330	1.465	1.925	2.215	2.775	2.479	0.601
Our proposed Method													
LessonPlanLM-7B	2.990	2.760	2.690	2.650	2.640	2.965	2.140	2.230	2.582	2.760	2.505	2.709	0.665
LessonPlanLM-72B	**3.000**	**2.780**	**2.930**	**2.845**	**2.655**	**3.000**	**2.565**	**2.735**	**2.787**	**2.775**	**2.810**	**2.843**	0.816

The best results are highlighted in bold.

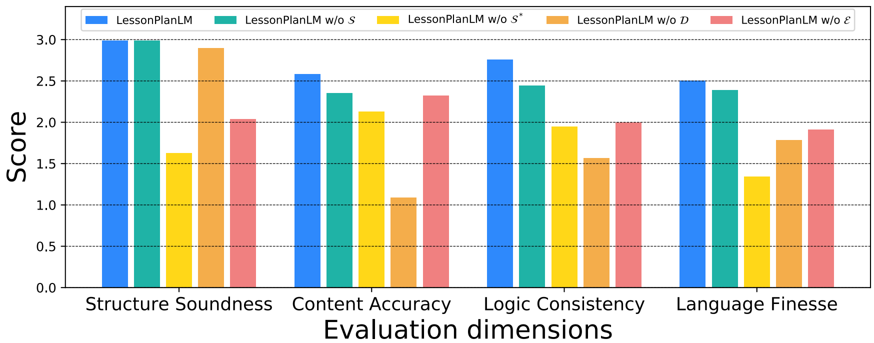

Fig. 5 The performance of different variants in LessonPlanLM.

Table 5 The influence of RAFT in LessonPlanLM.

Method	Structure Soundness	Content Accuracy	Logic Consistency	Language Finesse
LessonPlanLM	2.990	2.581	2.760	2.505
LessonPlanLM w/o RAFT	2.990	2.144	2.500	2.060

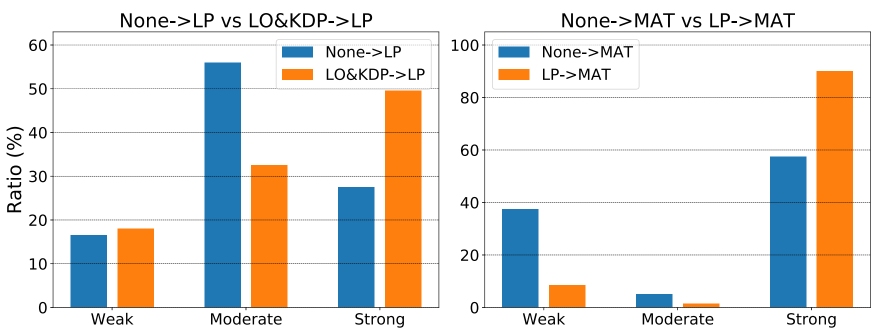

Fig. 6 The influence of consistent consideration of lesson plan components.

Future work

While LessonPlanLM currently focuses on the lesson plan paradigm within the Chinese education system, our evaluation metric is designed with generalizable principles that align with widely accepted pedagogical standards. Moreover, our framework already supports lesson plan generation in English. In future work, we plan to extend LessonPlanLM to accommodate diverse curricular structures and lesson plan formats across different countries and educational systems. By systematically investigating regional teaching standards and instructional requirements, we aim to make LessonPlanLM a globally applicable tool that can effectively support educators worldwide and reduce their lesson planning workload.

Table 6 LessonPlanLM vs. Human.

	LessonPlanLM vs. NPSSSEP/Wookey			
	Win (%)	**Tie (%)**	**Loss (%)**	**Kappa**
Structure Soundness	44.00/47.62	55.33/52.38	0.67/0.00	0.486/0.647
Content Accuracy	42.02/39.87	39.18/48.07	18.80/12.06	0.487/0.669
Logic Consistency	36.11/25.93	55.56/74.07	8.33/0.00	0.552/0.818
Language Finesse	8.67/0.67	84.00/87.33	7.33/12.00	0.479/0.942

Prompt for Unifying the Format of Lesson Plans by GPT-4

Please follow the {output format} to format the {component} section of a lesson plan.

Prompt for Rewriting Search Results by GPT-4

You are now an excellent information extraction and organization expert. I want to create a lesson plan with {teaching information}, but I lack detailed knowledge about it. By searching, I have gathered various {teaching information} components. Your task is to extract relevant information from these {search results} to help me complete the following components of a lesson plan: ["textbook analysis", "student learning analysis", "lesson objectives" and "key & difficult points"].

Prompt for Generating Simulated Lesson Plans by GPT-4

You are now an expert in the {subject} and have won multiple awards for instructional design. Your task is to write the corresponding content of the {component} section of a lesson plan with the provided teaching information. The teaching information is as follows:
Subject: {subject}
Textbook Version: {textbook version}
Grade: {grade}
Lesson Topic: {lesson topic}

This is an example:{example}

Please write the corresponding content of the{component} section with the given example.

Fig. 7 The prompts used in LessonPlanLM.

Data availability
The data that support the findings of this study are openly available at https://github.com/ai4ed/LessonPlan.

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Notes
1 https://paper.iflytek.com/
2 https://claude.ai
3 https://deepmind.google/technologies/gemini/
4 https://basic.smartedu.cn/
5 https://www.wookey.cn/

Appendix

Prompts used in LessonPlanLM
The prompts for GPT-4 to format all semantic components within each component into a uniform structure, rewrite the search results to construct knowledge retriever $\mathcal{E}$ and use the lesson plan semantic components $\mathcal{S}$ to generate simulated lesson plans $\mathcal{S}^*$ are illustrated in Fig. 7.

Prompts for lesson plan generation of baselines
In this work, we design a series of prompts to effectively guide existing LLMs for automatic lesson plan generation. With comprehensive experimentation, we observe that LLMs such as ERNIE Bot 4.0, iFlytek Spark 3.5 and GLM-4 exhibit certain limitations in understanding the concept of lesson plan modules. Consequently, when designing prompts for these models, we generate complete lesson plans directly. However, during the process of using GPT-4, Claude 3.5 connect and Gemini 1.5 Pro to generate lesson plan data, we create individual prompts for each component to enhance the quality and stability of the generated outputs. The details of these prompts are shown in Fig. 8.

Comparisons of automatic evaluation and human evaluation
Table 7 reports the automatic and human evaluation results of lesson plans generated by ERNIE Bot 4.0, iFlytek Spark 3.5, GLM-4 and our LessonPlanLM. We can see that there are large scoring gaps between GPT-4 and human evaluation. We suggest that

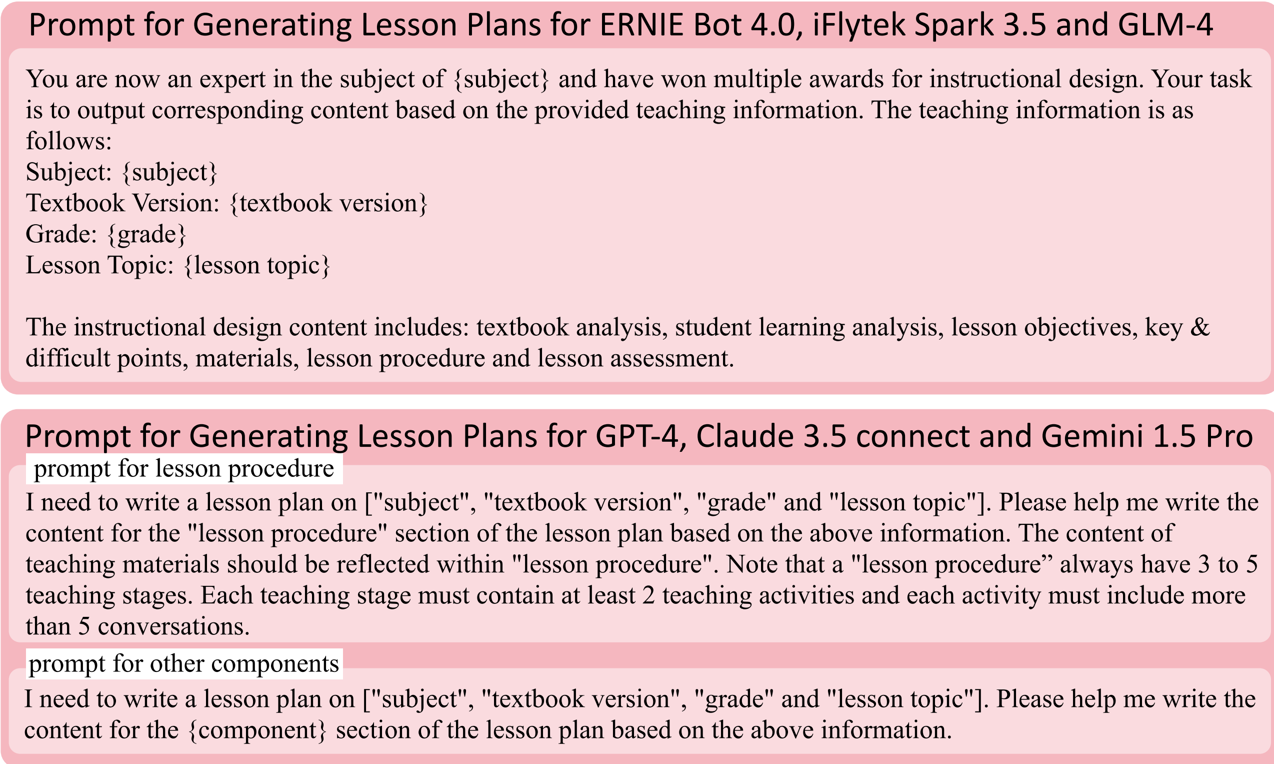

Fig. 8 The prompts for generating lesson plans by existing LLMs.

Table 7 The automatic and human evaluation results of lesson plan generation performance.

Method	Annotator	Structure Soundness	Content Accuracy								Logic Consistency	Language Finesse	Overall
			TA	SLA	LO	KDP	MAT	LP	AE	AVG.			
ERNIE Bot 4.0	Human	2.995	1.700	1.900	2.275	1.800	1.800	1.300	2.000	1.825	1.800	2.040	2.165
	GPT-4		2.800	2.475	3.000	3.000	2.275	2.355	2.645	2.650	1.335	2.425	2.137
iFlytek Spark 3.5	Human	2.910	1.010	1.020	1.700	1.540	1.040	1.190	1.020	1.217	2.745	1.035	1.977
	GPT-4		2.145	2.300	2.985	2.995	1.675	2.140	1.865	2.301	1.265	1.581	1.716
GLM-4	Human	2.195	1.000	1.070	2.230	1.020	1.030	1.210	2.060	1.374	2.040	1.080	1.672
	GPT-4		2.540	2.675	2.965	2.975	2.415	2.160	2.935	2.666	1.335	2.205	2.069
LessonPlanLM-7B	Human	2.990	2.760	2.690	2.650	2.640	2.965	2.140	2.230	2.582	2.760	2.505	2.709
	GPT-4		2.310	2.955	3.000	2.995	2.890	2.430	2.990	2.796	1.340	2.110	2.082
LessonPlanLM-72B	Human	3.000	2.780	2.930	2.845	2.655	3.000	2.565	2.735	2.787	2.775	2.810	2.843
	GPT-4		2.270	2.965	2.950	2.980	2.945	2.545	3.000	2.808	1.425	2.275	2.169

GPT-4 cannot complete our scoring task, we only use human evaluations to assess lesson plans.

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Author contributions
YZ. SH, XZ, YH wrote the main manuscript text. SH processed the data. YZ and XZ conducted experimental analysis. YH designed the evaluation metrics. ZL and WL proposed the main idea and revised the paper. All authors reviewed the manuscript.

Competing interests
The authors declare no competing interests.

Ethical approval
Ethical approval was not required as the study did not involve human participants.

Informed consent
Informed consent was not required as the study did not involve human participants.

Additional information
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